



Society of Petroleum Engineers

SPE-229638-MS

Sustainable Drilling Using Batch/Pad Drilling: A Strategic Approach for Reduced Environmental Impact

A. H. Abd El Aziz, M. Mostafa, G. Arbadov, A. Popov, A. Ibrahim, M. Anwar, A. Alajami, and E. Al Shamisi, ADNOC Onshore

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229638-MS](https://doi.org/10.2118/229638-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

As the energy industry transitions toward sustainable operations, batch and pad drilling have emerged as key strategies to minimize environmental impact while maintaining efficiency. This paper examines how these techniques significantly reduce water usage, chemical consumption, and CO₂ emissions—aligning with The operating company's sustainability and operational optimization objectives.

By drilling multiple wells from a single location, this method decreases the need for repeated mobilization and demobilization of equipment, leading to lower fuel consumption and greenhouse gas emissions. Additionally, centralized water and chemical management in pad drilling reduces waste and optimizes resource utilization. Advanced technologies such as closed-loop drilling systems and real-time monitoring further enhance sustainability by minimizing fluid loss and improving drilling accuracy. The integration of digital technologies and advanced planning supports The operating company's broader strategy of decarbonization and resource optimization.

Case studies from UAE unconventional reservoirs demonstrate that pad drilling not only supports Net Zero 2050 targets but also improves cost efficiency through reduced downtime and streamlined logistics. Drilling fluid management systems enable 30–50% less freshwater consumption through recycling and closed-loop fluids, while batch chemical handling minimizes waste and improves treatment efficiency. These practices reduce fuel consumption and greenhouse gas emissions (e.g. CO₂) by up to ~40%, lowering the overall carbon footprint of drilling activities. The adoption of advanced technologies—such as automated rigs, real-time monitoring, and AI-driven optimization—further enhances these benefits, ensuring precise drilling with lower resource intensity. This integrated approach not only supports environmental stewardship but also improves cost-effectiveness and operational consistency, marking a critical step toward greener energy production and resource optimization.

Through the adoption of batch/pad drilling, The operating company not only enhances operational performance and cost-effectiveness but also underscores its leadership in responsible energy production. This practice supports The operating company's framework, highlights how sustainable drilling can balance economic and environmental performance, and reinforces its role in driving sustainable development within

the UAE. It sets a benchmark for the industry and a precedent for environmentally conscious upstream operations across the region.

Keywords: Batch drilling, pad drilling, sustainability, CO₂ reduction, operational efficiency, closed-loop systems, decarbonization

Introduction

The global energy sector is undergoing a paradigm shift, with increasing emphasis on sustainable operations that balance economic viability with environmental responsibility, seeking to "produce more lower-cost and lower carbon-intensive barrels". In this context, batch and pad drilling have emerged as transformative methodologies that reduce ecological impact while enhancing operational efficiency.

The operating company has been at the forefront of adopting these techniques to align with the UAE's Net Zero 2050 Strategic Initiative. By drilling multiple wells from a single location, pad drilling minimizes repetitive mobilization and demobilization of equipment, leading to substantial reductions in fuel consumption, water usage, and chemical waste.

This paper explores the environmental and operational benefits of batch and pad drilling, supported by case studies from UAE unconventional reservoirs. It also highlights the role of advanced technologies—such as closed-loop drilling systems, real-time monitoring, and AI-driven optimization—in maximizing sustainability outcomes.

Batch/Pad Drilling Concepts

Conventional Drilling (Figure 1) is one well per location using conventional rig (Not skidding), Install the cellar, Drive and cement conductor with service company before Rig move to location. **Unconventional reservoirs** which requires hydraulic fracturing to produce hydrocarbon due to low reservoir permeability leading to low drainage area which requires more wells to reach the reservoir boundaries (Figure 2) and maximize the drainage area. **Pad drilling** (Elzeky et al., 2024) (Figure 3) uses a single drilling pad to handle several horizontally drilled wells in different directions. After drilling one well, the rig can be hydraulically "walked" or skidded a short distance to the next hole on the same pad, rather than fully rigging down and up between isolated locations. This cluster approach allows targeting large subsurface areas with fewer surface sites and faster rig moves.

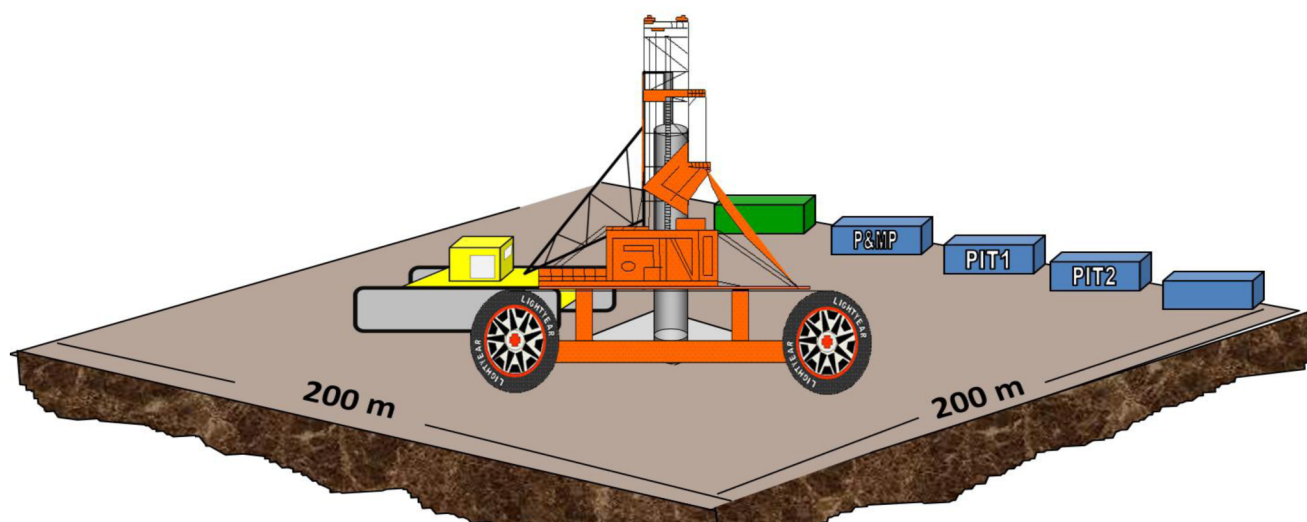


Figure 1—Conventional Drilling Well location

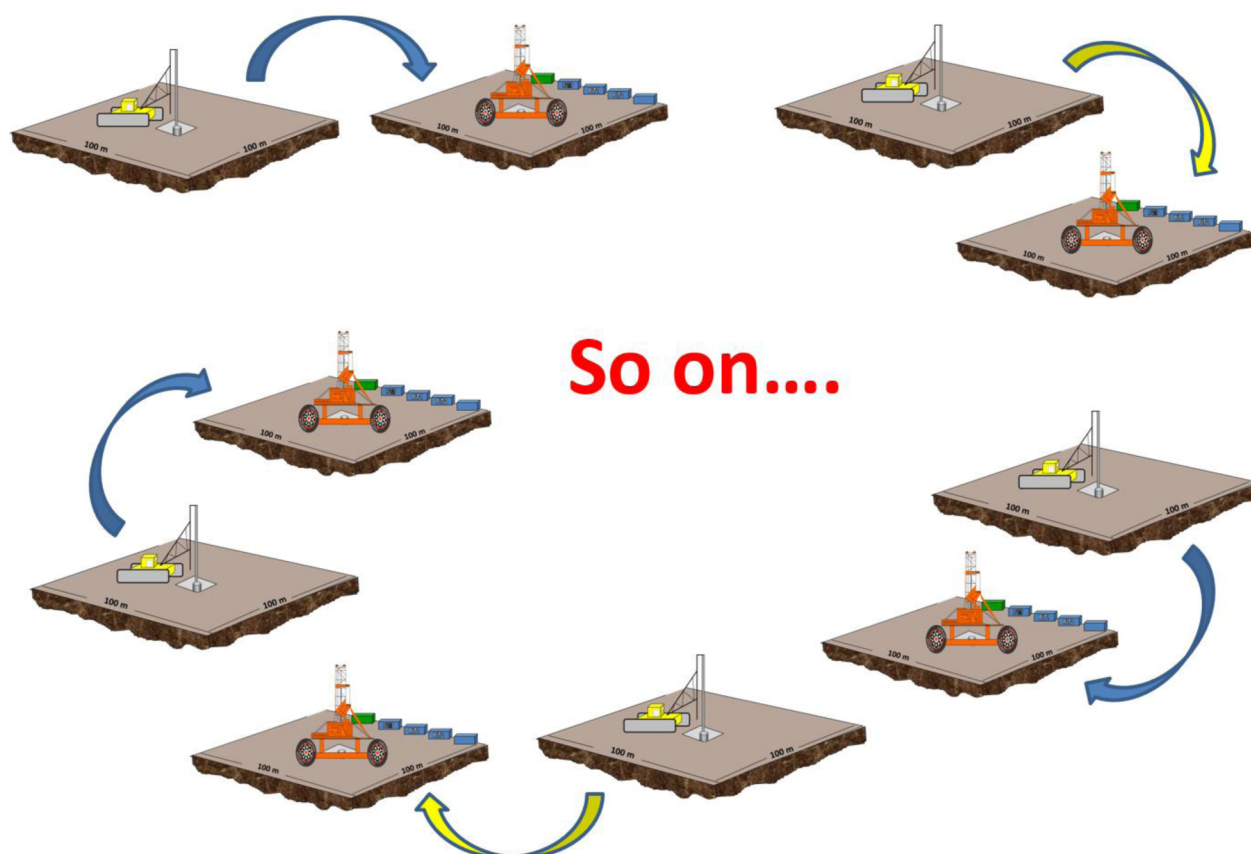


Figure 2—Conventional Drilling Wells to reach reservoir boundaries

Pad and batch drilling have become important for reducing the environmental footprint of well development. By concentrating operations at one site, pad drilling minimizes land disturbance and saves time. Pad drilling thus leverages geometry and sequencing to reduce equipment moves and non-productive time. As one industry summary notes, modern pad drilling "allows rig operators to drill groups of wells more efficiently" and concentrates wellheads to reduce above-ground management costs.

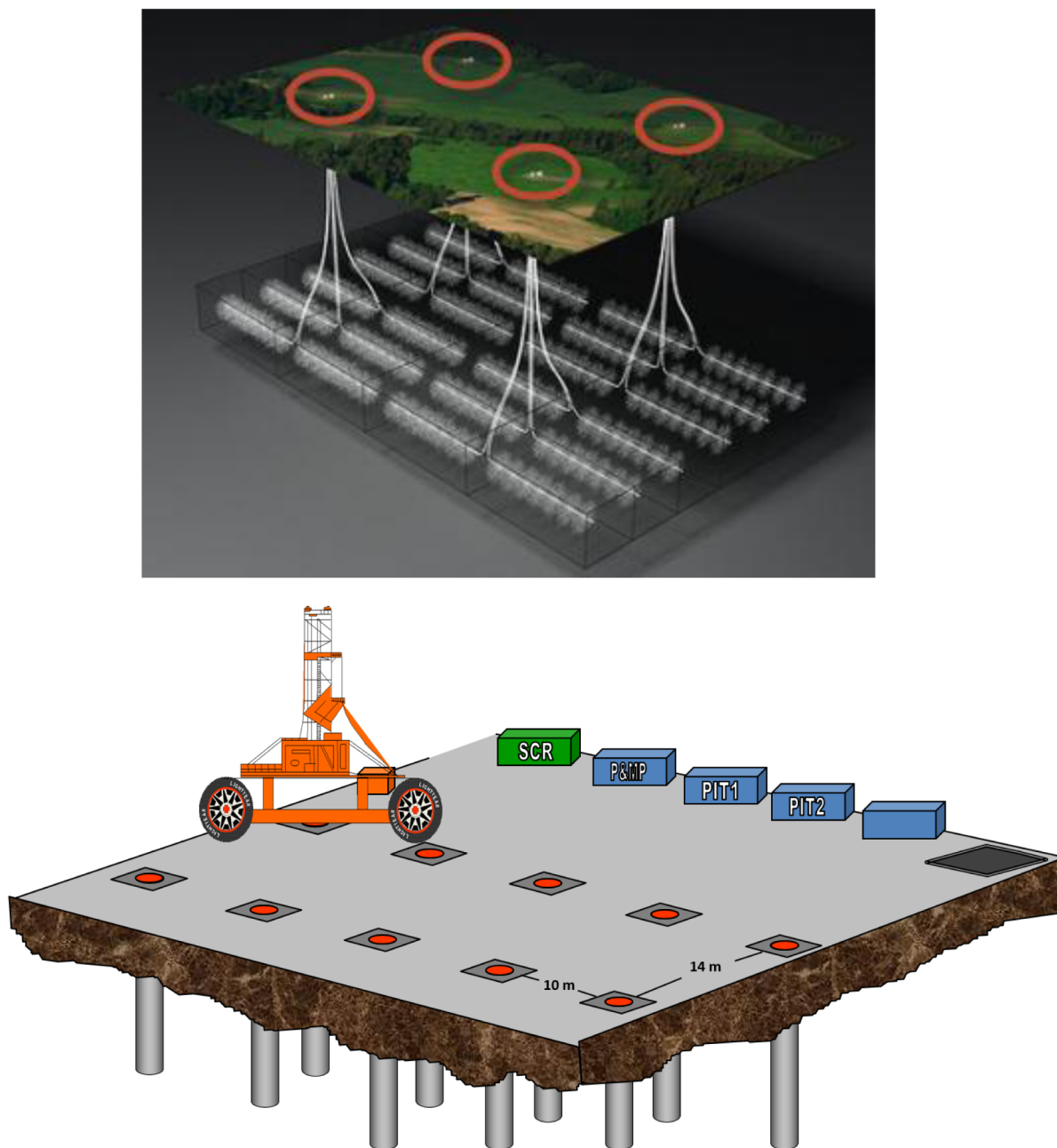


Figure 3—Drilling rig at a pad in an unconventional reservoir. Multi-well pad drilling concentrates wellheads and infrastructure, allowing horizontal wells in multiple directions to be drilled from one site, which minimizes land disturbance and rig move

Batch Drilling Operations: Within a pad, wells can be drilled in batches. For example, an unconventional pad might consist of 10 wells drilled in 3 batches (Figure 4) or 2 (Figure 5) batches. Three batches design depend on drilling the surface section per each well then skid to drill intermediate section per each well after that drill production section and complete the well with frac string per each well. While Two Batches (Figure 5) drilling depend on combining the surface and intermediate section with each other per each well and after that drill the production section and complete the well with frac string per each well.

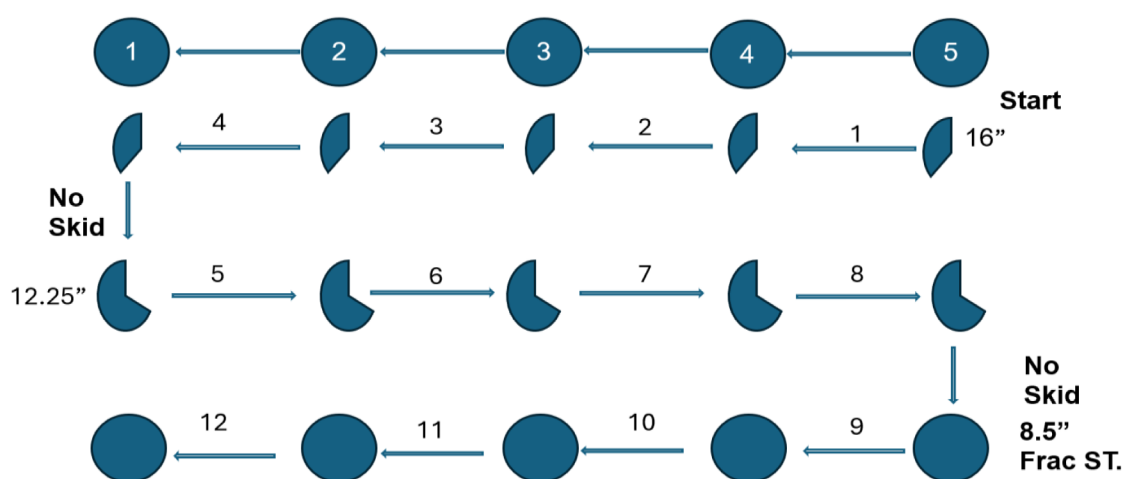


Figure 4—Batch/pad drilling skidding sequence in 3 batches

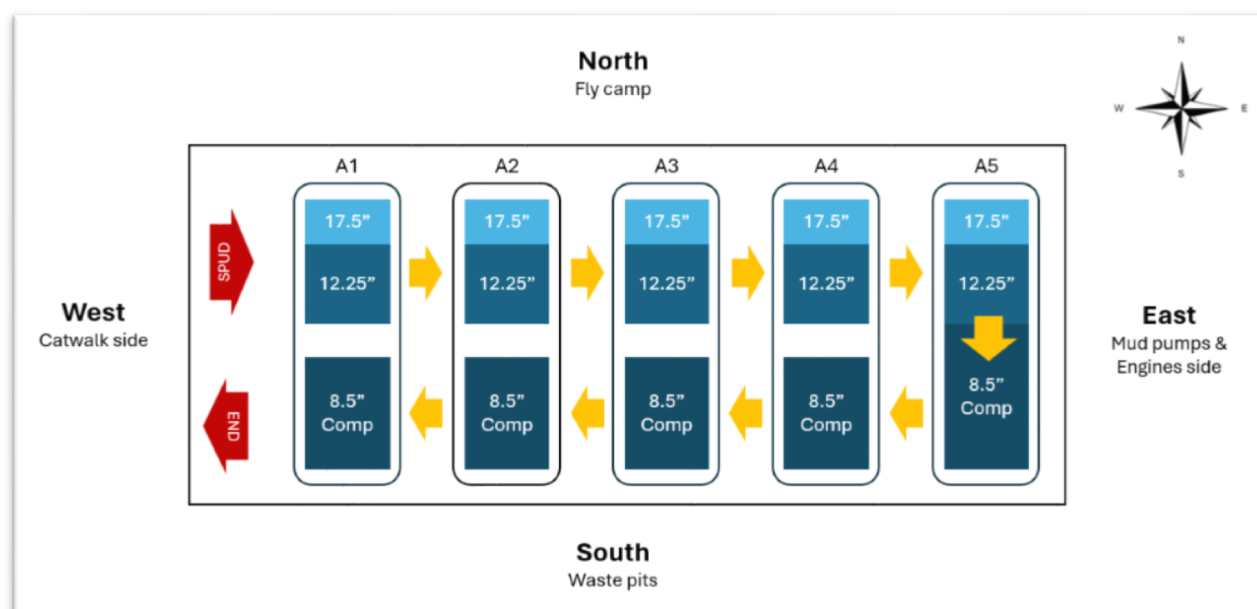


Figure 5—Batch/pad drilling skidding sequence in 2 batches

The criteria for 3 batches (Figure 4) drilling depend on having different mud type and mud weight per each section so it is better to drill section by section to minimize the dump and dilute operations for mud.

The criteria for 2 batches (Figure 5) drilling depend on minimizing the total number of skids as it is applicable to continue with the same mud in the intermediate section

Batch scheduling lets crews repeat similar operations in succession, applying lessons learned immediately and efficiently reusing materials like mud and casing. Also saving the flat time and minimize the invisible lost time (P/U M/U Drillpipe, M/U BHA, Nipple up riser/diverter/ BOP,R/U and lay down of Mud logging, directional sensors and unit, ...etc)

Sustainable Drilling Techniques

Advanced drilling methods complement pad drilling to further improve sustainability. Key techniques include:

- **Environmental impact and Footprint:** Ecological living aims to live in respect with nature and living things by avoiding waste. By reducing consumption and waste production as much as possible, through minimizing the oil and gas wells location as one location serves for more than 5 wells, which minimize the environmental impact. Moreover, the land disturbance per well is dramatically lower. Using one pad instead of many distributed well sites preserves habitat and eases reclamation. Surface disturbance is cut by drilling multiple wells on a smaller area. Companies have demonstrated drilling up to four wells on a single pad of about one acre, versus four separate pads for conventional wells. This means less road-building, less clearing, and lower soil erosion. Minimize the CO₂ emission by more than 40% related to move equipment and location preparation equipment. Drilling wastes have been collected in one location instead of more than 5 locations which will ease the safe disposal of drilling wastes.
- **Well delivery duration and Cost impact:** By minimizing the invisible lost time by 20-30%, reuse of mud chemicals, minimizing of water and chemical consumptions, efficient drilling, decreasing the move cost and move duration, decreasing the waste management cost, improve the drilling performance due to minimizing the drilling uncertainties and improve the learning curve all these factors contribute to minimize the drilling cost and minimize the well delivery duration as it dropped more than 67% less than average well duration. (Riyami et al., 2016; Ruzhnikov, 2025; Al-Ameri et al., 2022)
- **Closed-Loop Fluid Management:** Closed-loop drilling systems recycle drilling fluids and cuttings instead of dumping into open pits. These systems "capture and recycle drilling fluids, minimizing water consumption and preventing the release of pollutants". These systems recycle drilling fluids, reducing freshwater demand by 30–50%. By eliminating reserve pits, operators can reuse drilling mud for multiple wells and "minimize cuttings volume and waste disposal ". This reduces the need for fresh water and decreases waste disposal costs. Closed-loop rigs, therefore, align with sustainability by cutting both water use and waste. (Smith, J. et al., 2022)
- **Batch Chemical Handling:** Similarly, pad-based operations enable central handling of additives and chemicals. By batching wells, chemical inventory can be shared and mixing/treatment done on-site in bulk, reducing chemical waste. (While specific data for batch chemical savings are scarce in the literature, operators report that planned, centralized chemical mixing on pads improves treatment efficiency and minimizes leftover waste.)
- **Directional/Horizontal Drilling:** As noted above, drilling horizontal wells from a single pad leverages directional technologies to hit multiple targets without moving rigs. This not only reduces surface footprint but also concentrates drill site infrastructure, lowering overall land use and permitting burden.
- **Real-Time Monitoring and AI:** Digital technologies are deployed to optimize drilling in real-time. Systems for downhole monitoring and data analytics help detect issues before they cause waste or downtime. Minimizing the wells separation distance will minimize the uncertainties between the wells which help for better simulation and Better detecting of drilling anomalies and dealing with same. By integrating digital planning and execution tools, operators can continuously adjust drilling parameters to minimize energy use and emissions per foot drilled. For example, AI-driven workflows and digital twins can simulate drilling scenarios to choose the least-energy-intensive approaches. In pad operations, real-time data sharing across wells further boosts efficiency.
- **Automated and Novel Drilling Hardware:** Advanced hardware like autonomous walking rigs and at-bit steering systems reduce the number of trips needed. Eliminating the need for BHA reconfiguration" and permits drilling vertical, curve, and lateral sections in one run. This "reduces energy consumption and emissions" by cutting down trips. In fact, more sections can be drilled in one run via same tools which lowers CO₂ emissions and make up and lay down time. Robotic pipe handling (e.g. automated catwalks) similarly speeds up tripping and reduces personnel exposure

to hazards. These digital tools are crucial for the decarbonization push: the digitalization must now "go hand in hand with decarbonization" to reduce emissions per foot drilled. By aligning drilling KPIs with carbon metrics in planning software, operators can minimize Scope 1 emissions. Likewise, emissions-tracking platforms provide transparency during pad operations, letting drillers tweak engine loading and fuel use in real time.(Borozdin et al., 2020; Duplyakin & Kapranov, 2019)

- **Waste Minimization:** By reusing mud and cuttings, pad drilling cuts waste volumes. Drill cuttings can be reinjected or repurposed rather than landfilled, and any remaining waste is highly consolidated on fewer pads. This streamlines waste management and reduces chemical additives needed per well.
- **Reduced Fuel and Emissions:** Fewer rig moves mean less fuel burned by trucks and rigs. Engineering studies and operations data confirm large efficiency gains. For instance, an Abu Dhabi pad project achieved a 53% reduction in average drilling cycle time for four pad wells, greatly cutting idle and transit time. Reduced rig mobilizations directly lower diesel consumption and CO₂ output. Pad drilling compounds this by cutting rig moves: one study noted a ~40% overall carbon-footprint reduction in pad wells, thanks to less mobilization (though exact figures depend on rig type and moves)(Gilbert & Getliff, 2010, Al Mansoori et al., 2023)
- **KPI and benchmarking the rig performance:** Invisible lost time and flat time in drilling operations represents more than 30% from the total well duration. By benchmarking each operations in the drilling process we can track the invisible lost time and minimize or eliminate the ILT. Example Weight to weight connection time has been decreased from 30.7 min to less than 4.9 min as shown in Figure 6

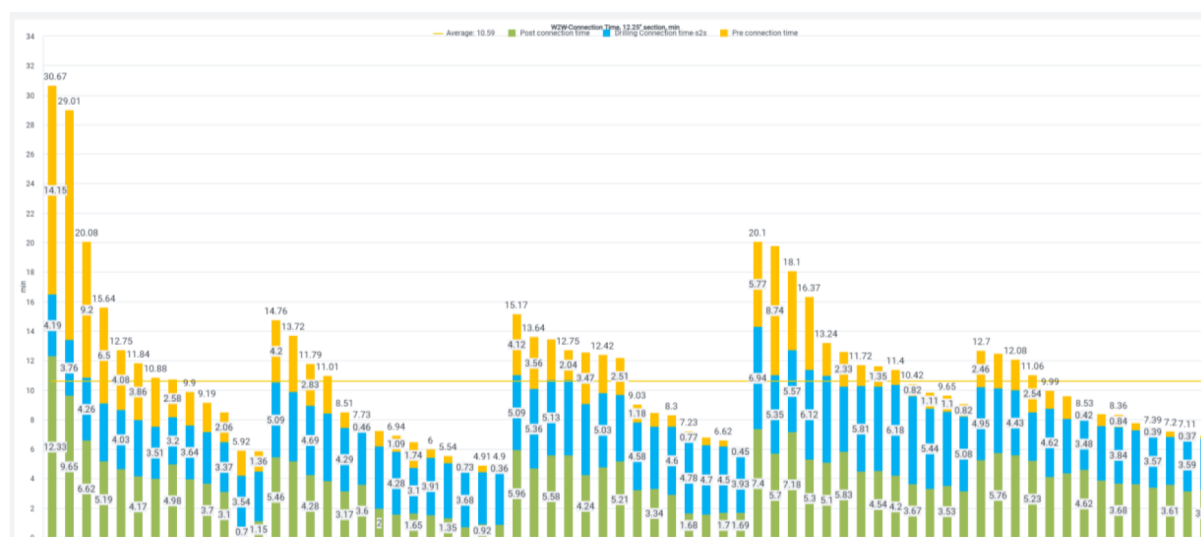


Figure 6—Oil Wells W2W Connection time for intermediate section

Source	Traditional Drilling	Pad Drilling	Reduction
Rig Moves	5-10 moves per project	1-2 moves	~50-60%
Fuel Consumption	5000-7000 gal/day	3000-4000 gal/day	~30-40%
CO ₂ Emissions	50-70 tons/well	30-40 tons/well	~30-40%

These combined effects are seen as critical enablers for The operating company's sustainable development. By demonstrating that "*sustainable drilling can balance economic and environmental performance*," The operating company positions itself as a leader. Their unconventional program's early

successes have set benchmarks for the region's energy transition, showing that high production targets and NetZero goals can be pursued in tandem.

Conclusions

Batch and pad drilling are transformative for sustainable upstream operations. By clustering wells and using advanced drilling systems, operators drastically cut downtime, resource use, and emissions. The operating company's application of these methods in its unconventional program has yielded concrete environmental metrics: major reductions in water and fuel use, sizable CO₂ cuts, and hundreds of millions in cost savings on large campaigns. When combined with closed-loop fluid systems, real-time optimization, and AI planning, pad drilling represents a holistic decarbonization strategy for drilling. Crucially, it proves that ambitious energy goals (NetZero2050) can align with operational performance goals.

This integrated approach is reproducible across oil and gas fields. It exemplifies how technology and planning underpin *responsible energy production*: smaller footprints, cleaner operations, and lower costs. As The operating company highlights, these gains underscore leadership in balancing economic growth with environmental stewardship. The UAE's experience sets an important precedent for the industry: multi-well pad drilling is a key practical pathway to a greener hydrocarbon future.

Acknowledgements

The authors thank The operating company, and associated field teams for access to operational data and support during field implementation phases. Appreciation is extended to digital analytics teams for providing real-time system optimization tools and technical expertise.

References

1. ADNOC. (2023). Sustainability Report 2023: Accelerating a Lower-Carbon Future. Abu Dhabi National Oil Company.
2. ADNOC. (2022). *Drilling Efficiency & Carbon Reduction Strategy: Roadmap to Net Zero 2050*. Abu Dhabi National Oil Company.
3. Elzeky, Moataz Mohamed Fahmy, Varma, Esha Narendra, Elhousseiny, Mahmoud, Anwar, Mohamed, Al Shamisi, Eisa Daban, and Ibrahim Hamdy. "Skidding Rig and Introduction of Batch Drilling System Increased Drilling Operation Efficiency in Onshore UAE Unconventional Project Saving 70mm \$." Paper presented at the APOGCE 2024, Perth, Australia, October 2024. doi: <https://doi.org/10.2118/221207-MSEL>
4. International Energy Agency (IEA). (2022). *The Role of Innovation in Reducing Upstream Emissions*.
5. SPE (Society of Petroleum Engineers). (2021). *Best Practices in Pad Drilling for Unconventional Reservoirs*. SPE-205432-MS. DOI: [10.2118/205432-MS](https://doi.org/10.2118/205432-MS)
6. Al-Mansoori, S., & Al-Hassani, H. (2023). "Carbon Footprint Reduction in UAE's Unconventional Reservoirs Through Batch Drilling." *Journal of Petroleum Science and Engineering*, **215**, 110702. DOI: [10.1016/j.petrol.2022.110702](https://doi.org/10.1016/j.petrol.2022.110702)
7. Smith, J., & Patel, R. (2022). "Closed-Loop Drilling Fluids: A Solution for Water Conservation in the Middle East." *Energy Reports*, **8**, 456–465. DOI: [10.1016/j.egy.2022.01.003](https://doi.org/10.1016/j.egy.2022.01.003)
8. Khan, M., & Al-Shehhi, M. (2021). "AI-Driven Optimization in Drilling Operations: A Case Study from ADNOC's Digital Transformation." *SPE Drilling & Completion*, **36**(3), 567–578. DOI: [10.2118/204567-PA](https://doi.org/10.2118/204567-PA)
9. World Bank. (2021). *Reducing Flaring and Methane Emissions in Oil & Gas Operations*.
10. Available: <https://www.worldbank.org/en/topic/extractiveindustries>

11. McKinsey & Company. (2020). *The Future of Drilling: Automation and Sustainability*.
12. Available: <https://www.mckinsey.com/industries/oil-and-gas>
13. Al-Ameri, W., & Al-Nuaimi, K. (2023). "Economic and Environmental Benefits of Multi-Well Pad Drilling in the Middle East." *International Journal of Oil, Gas and Coal Technology*, **32**(2), 145–160. DOI: [10.1504/IJOGCT.2023.10052145](https://doi.org/10.1504/IJOGCT.2023.10052145)
14. Gupta, A., & Zhang, Y. (2022). "Real-Time Drilling Monitoring Using IoT and Machine Learning." *Journal of Natural Gas Science and Engineering*, **98**, 104356. DOI: [10.1016/j.jngse.2021.104356](https://doi.org/10.1016/j.jngse.2021.104356)
15. Abbas, A. K., Assi, A. H., Abbas, H., Almubarak, H., & Al Saba, M. 2019. Drill Bit Selection Optimization Based on Rate of Penetration: Application of Artificial Neural Networks and Genetic Algorithms. Presented at the Society of Petroleum Engineers Annual Technical Conference and Exhibition. SPE 197241 MS. <https://doi.org/10.2118/197241MS>
16. Alkinani, H. H., Al Hameedi, A. T. T., Dunn Norman, S., Alkhamis, M. M., & Mutar, R. A. 2019. Prediction of Lost Circulation Prior to Drilling for Induced Fractures Formations Using Artificial Neural Networks. Presented at SPE Oklahoma City Oil and Gas Symposium, Oklahoma City, Oklahoma, USA, 9–10 April. SPE 195197 MS. <https://doi.org/10.2118/195197MS>
17. Borozdin, S., Semenova Chashchina, O. K., Fitsner, L. K., et al 2020. Drilling Problems Forecast System Based on Neural Network. Presented at the SPE Annual Caspian Technical Conference, virtual, 21–22 Oct. SPE 202546 MS. <https://doi.org/10.2118/202546MS>
18. Dmitrievsky, A. N., Duplyakin, V. O., Eremin, N. A., & Kapranov, V. V. 2019. Algorithm for Creating a Neural Network Model for Classification in Systems for Preventing Complications and Emergencies in Construction of Oil and Gas Wells. *Sensors & Systems (Russia)*, **243**(12): 3–10.
19. Lind, Y. B., & Kabirova, A. R. 2014. Artificial Neural Networks in Drilling Troubles Prediction. Presented at SPE Russian Oil & Gas Exploration and Production Technical Conference and Exhibition, Moscow, 14–16 Oct. SPE 171274 MS. <https://doi.org/10.2118/171274MS>
20. Macpherson, J. D., de Wardt, J. P., Florence, F. et al 2013. Drilling Systems Automation: Current State, Initiatives, and Weakley. Presented at Offshore Technology Conference, Houston, Texas, USA. OTC 25838. <https://doi.org/10.4043/25838MS>
21. Verret, J., et al (2016). Operational Excellence in Unconventional Drilling Using Batch and Pad Strategies. SPE 180858.
22. Al Gharbi, A., & Al Ameri, A. (2021). Optimizing Unconventional Well Delivery in UAE Onshore Fields Using Integrated Drilling Execution Platforms. SPE 204132.
23. Younis, M., et al (2019). Reducing Carbon Footprint Through Efficient Rig Moves and Batch Drilling Techniques. SPE 195566.
24. Mahmoud, M., et al (2023). Closed-Loop Drilling and AI-Based Optimization for Minimizing Drilling Waste and Emissions in Unconventional Wells. SPE 212445.
25. IEA. (2021). Drilling and Emissions Reduction Strategies in the Upstream Sector. International Energy Agency.
26. ADNOC. (2023). Sustainability Report 2023: Decarbonizing Drilling and Emissions Reduction through Operational Efficiency.
27. Khan, R., et al (2021). Real-Time Drilling Emissions Monitoring and Performance Optimization. SPE 203380.
28. API. (2020). API RP 96: Deep Shale Well Design and Construction Guidelines. American Petroleum Institute.
29. Patel, A., et al (2020). Water Management and Environmental Optimization in Multi-Well Pad Drilling. SPE 198315.

30. UNEP. (2022). Best Environmental Practices for Upstream Oil and Gas Operations. United Nations Environment Programme
31. World Economic Forum. 2022. Oil and Gas Industry: A Net Zero Pathway. WEF.
32. ADNOC. 2022. ADNOC's Pathway to Net Zero by 2050. ADNOC Group Publications.
33. ADNOC. 2023. ADNOC Sustainability Report. Abu Dhabi National Oil Company.
34. Schlumberger. 2022. Closed Loop Drilling Systems: Enhancing Efficiency and Reducing Environmental Footprint. Technical Report.
35. Halliburton. 2021. Sustainable Drilling Solutions: Innovations in Fluids, Waste Management, and Rig Efficiency.
36. Society of Petroleum Engineers (SPE). 2020. Pad Drilling Strategy for Unconventional Reservoirs in the Middle East: Optimization, Challenges, and Results. SPE 199020 MS.
37. International Association of Oil & Gas Producers (IOGP). 2021. Best Practices in Wellsite Environmental Management. IOGP Report 543.
38. U.S. Department of Energy (DOE). 2020. Reducing Environmental Impact through Advanced Drilling Technologies.
39. ExxonMobil. 2019. Improving Efficiency Through Pad Drilling.
40. Ma, T., Chen, P., & Zhao, J. 2016. Overview on Vertical and Directional Drilling Technologies for the Exploration and Exploitation of Deep Petroleum Resources. *Geomechanics and Geophysics for Geo-Energy and Geo-Resources*, 2:365–395. <https://doi.org/10.1007/s409480160038>
[researchgate.net/sti.govlink.springer.com](https://www.researchgate.net/sti.govlink.springer.com)
41. Schlumberger. (2023). *Automated Drilling Rigs: Efficiency and Emission Reductions*.
42. API (American Petroleum Institute) *Guidelines on Sustainable Drilling Practices*.
43. IPIECA (International Petroleum Industry Environmental Conservation Association) reports on water management in drilling.